Supplement

Single-Platform Nanopore Sequencing Enables Diploid Telomere-to-Telomere Genome Assembly and Haplotype-Resolved 3D Chromatin Maps

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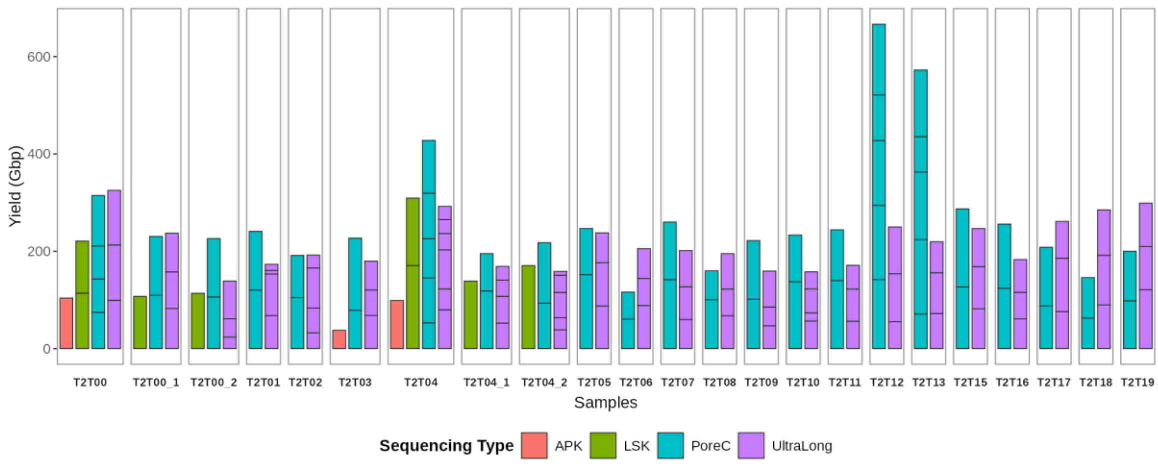
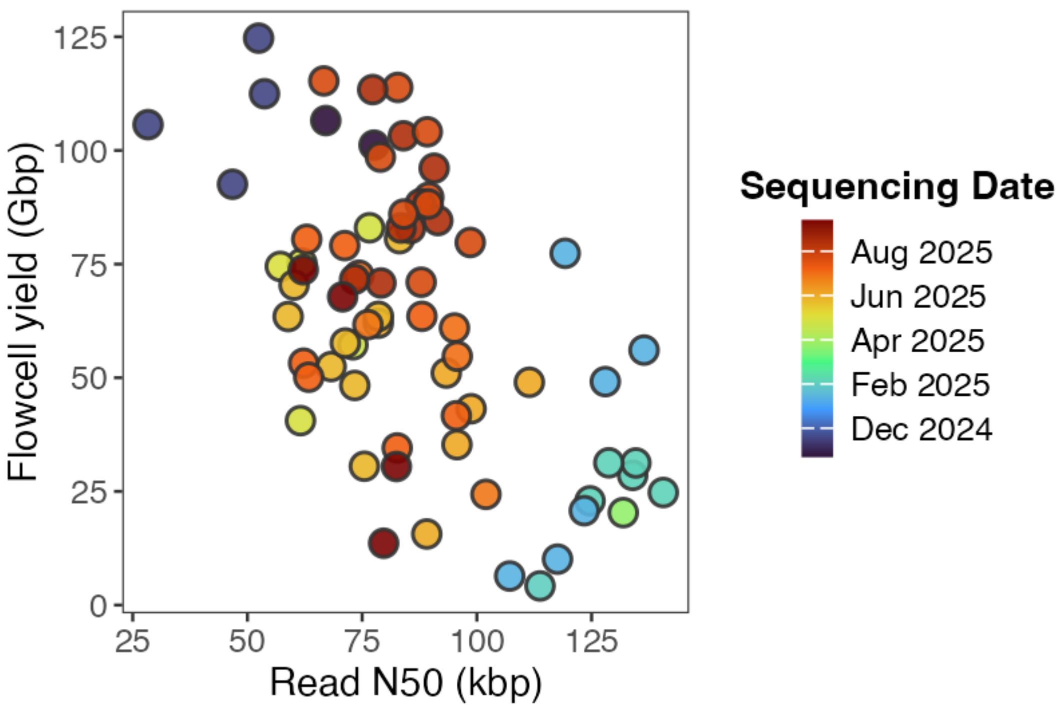
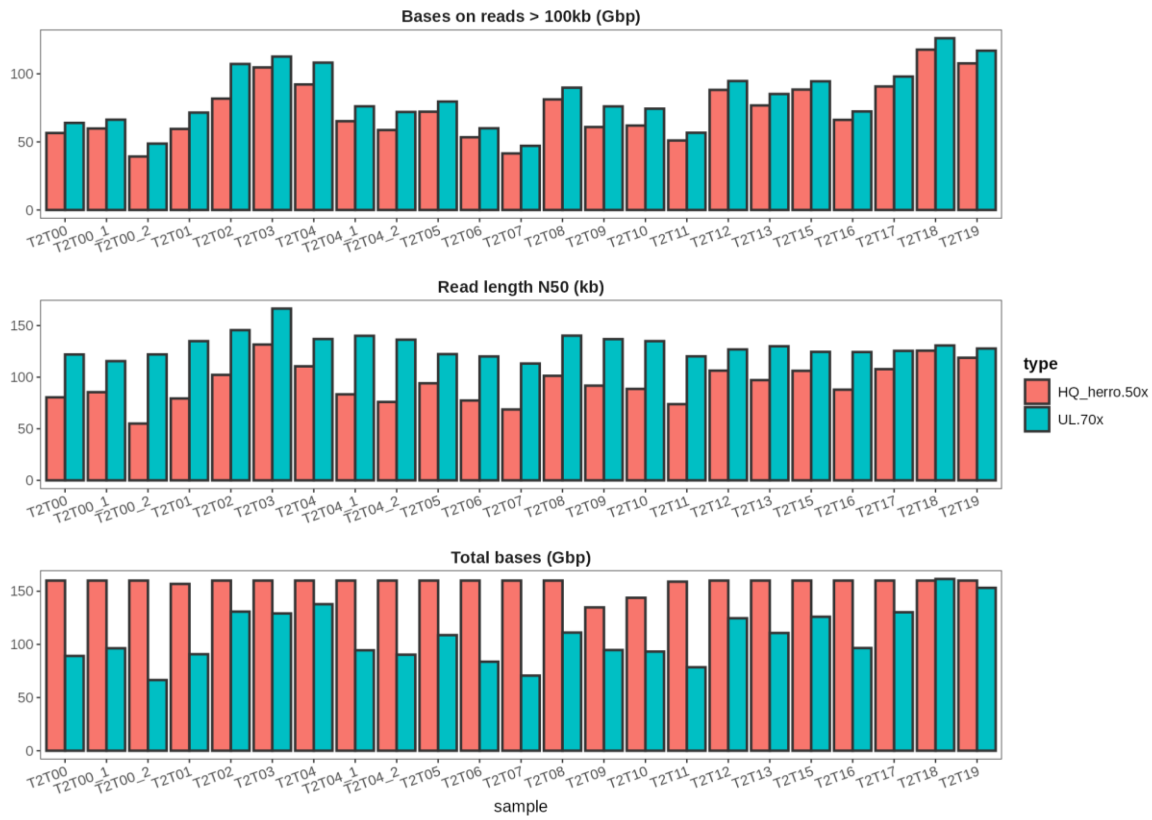


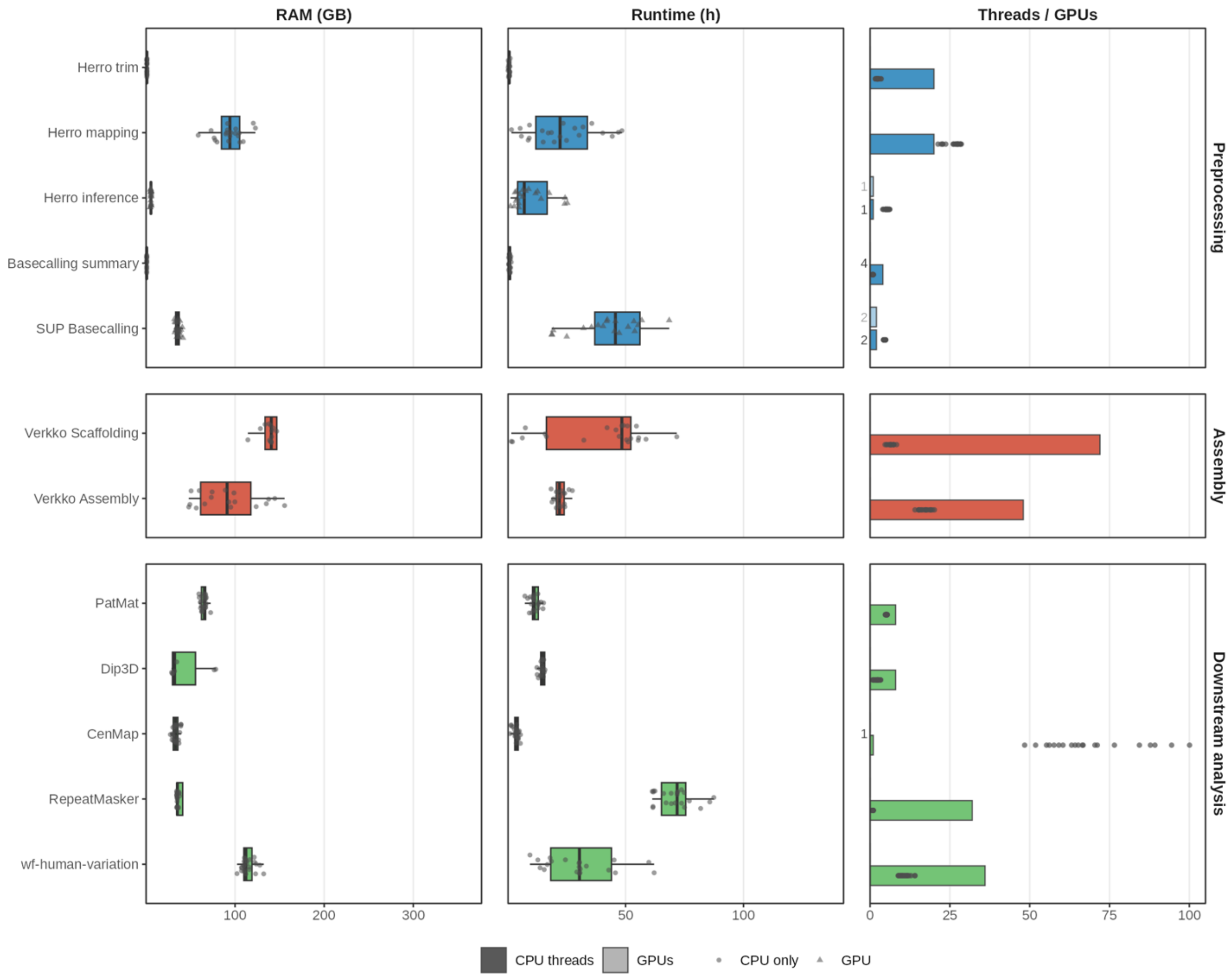
Supplementary Figure 1: Sequencing yield across all ONT flow cells. *Total sequencing output per flow cell for all samples. The standard workflow used three ultra-long (UL) and one Pore-C flow cell per individual for the assembly. Additional UL flow cells were sequenced for selected samples to compensate for low-yield runs. A single Pore-C flow cell was sufficient for assembly and haplotype phasing in all samples; a second Pore-C flow cell was sequenced for every sample to increase contact density for chromatin analyses, and additional Pore-C flow cells were generated for samples T2T00, T2T04, T2T12, and T2T13 to resolve contacts at higher resolution. For trio parents, one standard-length ligation sequencing (LSK) flow cell was added to supplement the HQ dataset. Assembly Polishing Kit (APK) was evaluated for reference-free consensus accuracy.*



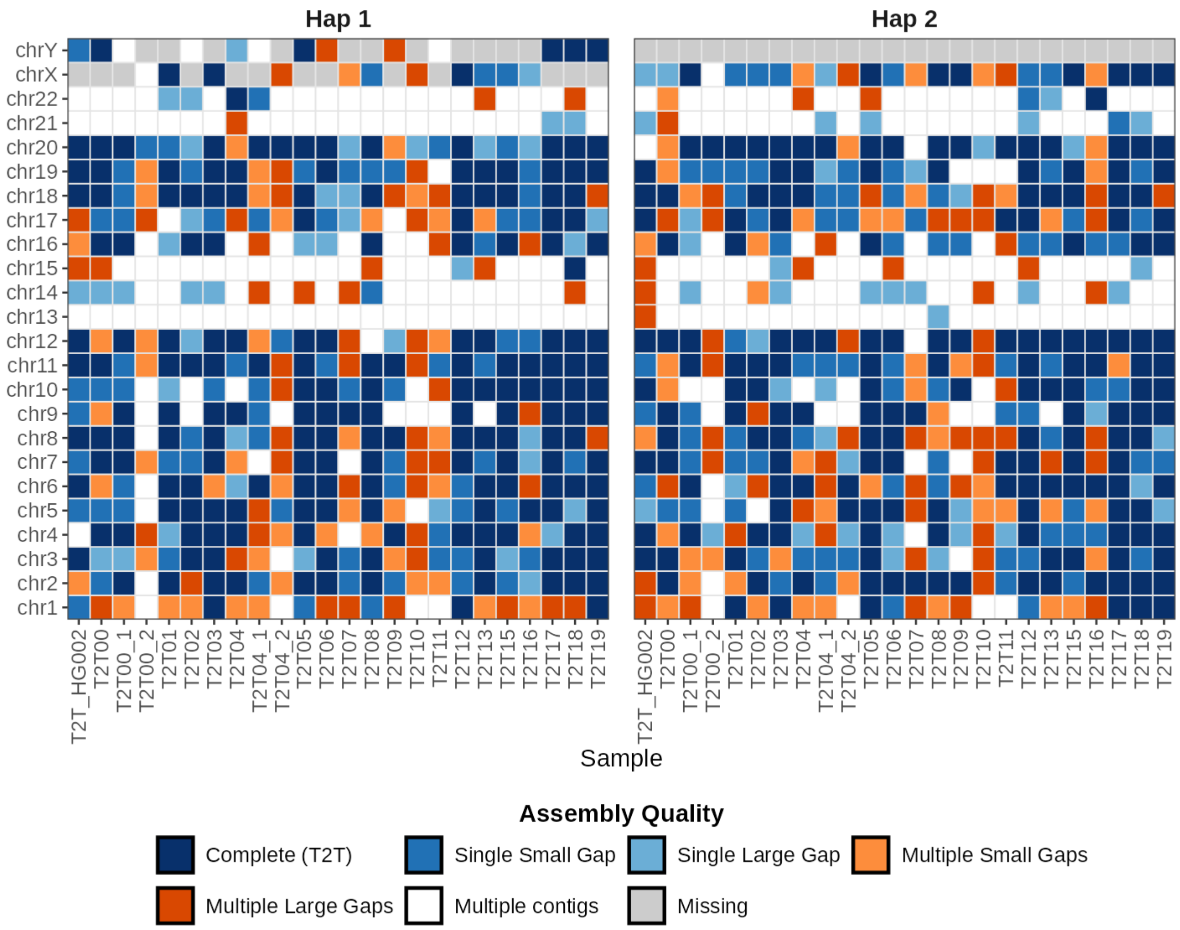
Supplementary Figure 2: Optimization of UL read length distribution. *Read length distributions of UL flow cells over the course of the project. Color scale reflects sequencing date. Progressive protocol optimization improved the balance between total yield and read length N50, increasing the proportion of reads >100 kb in later samples.*



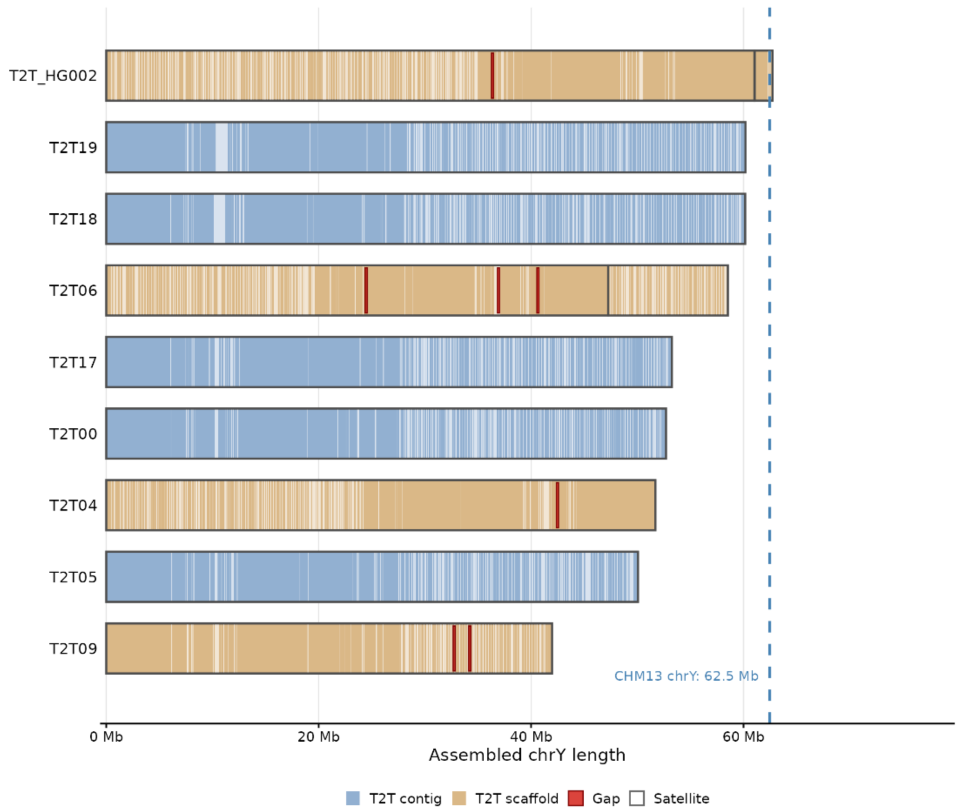
Supplementary Figure 3. Assembly input read statistics. *Read statistics for combined HQ and UL datasets used for assembly. HQ reads were capped at 50× coverage; UL reads were capped at 70× coverage and filtered to retain only reads ≥80 kb. Length-based subsampling prioritized longer reads to maximize repeat resolution.*



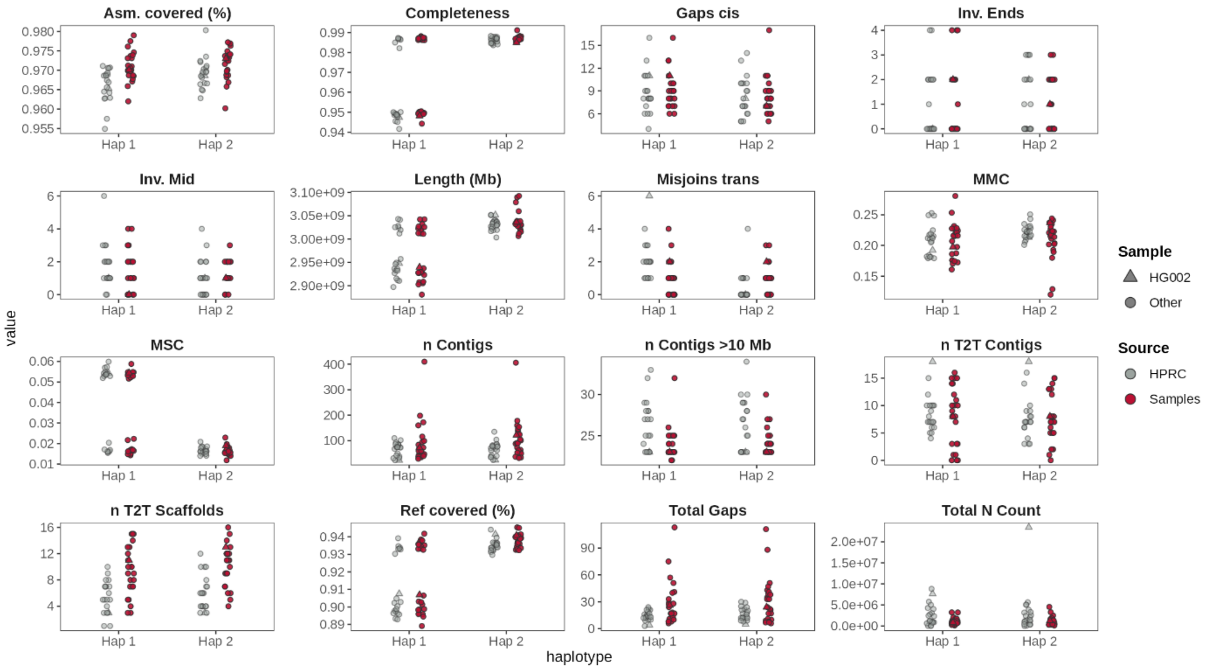
***Supplementary Figure 4. Computational resource requirements of the T2T-ONT assembly pipeline**. Runtime and resource requirements for the assembly pipeline steps running on a single server node with AMD Epyc and 2x Nvidia H100 GPU. Note 1: The high variation in the Verkko scaffolding runtimes is caused by partial caching of preexisting results in some samples, i.e. the higher values better reflect true runtime. Note 2: The CPU thread limitation for CenMap was not correctly configured, resulting in an unnecessary high allocation of cores during execution. This does not affect the correctness of results.*



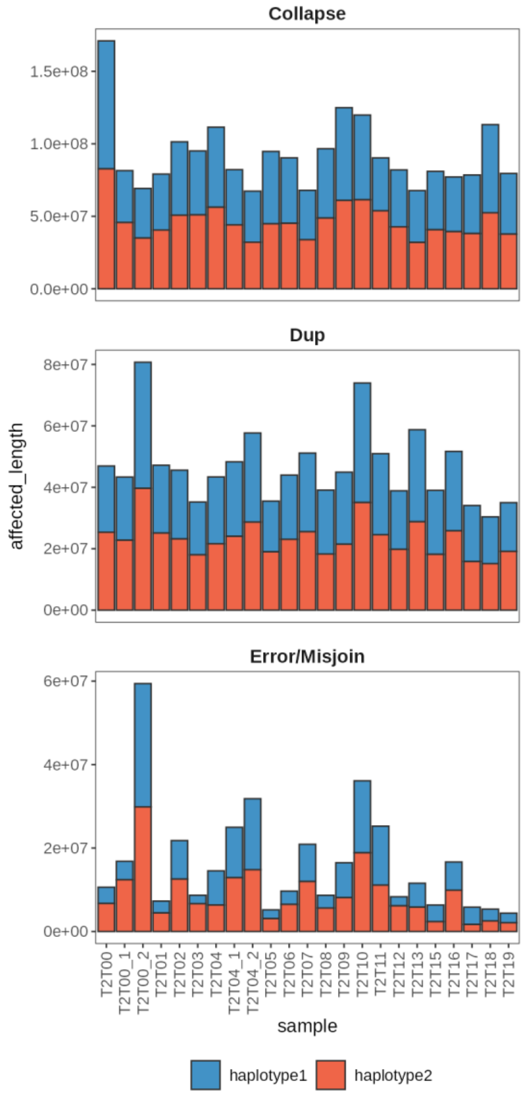
Supplementary Figure 5. Detailed chromosome assembly classification. *Chromosome-level assembly status for all haplotypes with expanded categorization. Dark blue indicates gapless T2T contigs spanning entire chromosomes. Light-colored categories represent T2T scaffolds containing gaps. Large gaps are defined as single gaps >50 kb or cumulative gap size >100 kb per chromosome. Sex chromosomes missing in the samples are shown with grey background.*



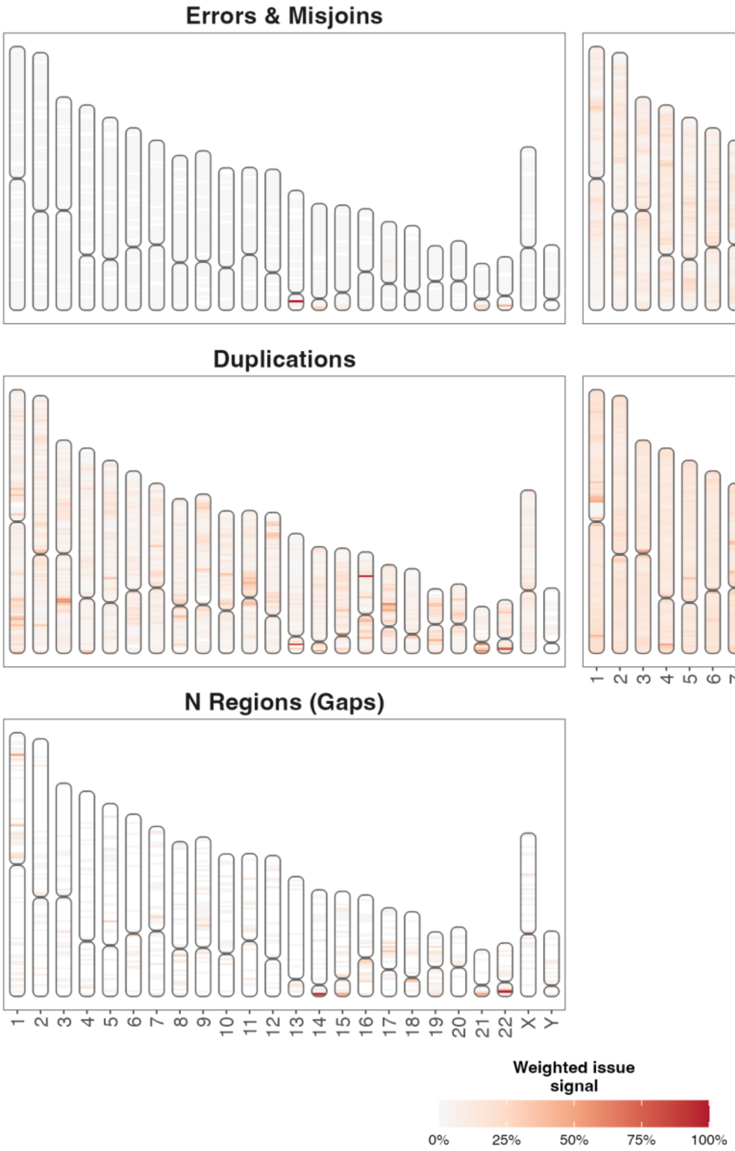
Supplementary Figure 6. Lengths of assembled T2T-resolved Y chromosomes. *Assembled chrY length for all male samples in which chromosome Y reached T2T status, colored by classification (T2T contig, T2T scaffold), with gaps and satellite regions marked. The dashed line marks CHM13v2 chrY (62.5 Mb). Assembled lengths range from about 42 to 61 Mb. Classification is based on a colinear alignment chain spanning at least 95% of both the assembled sequence and the reference chromosome (see Methods).*



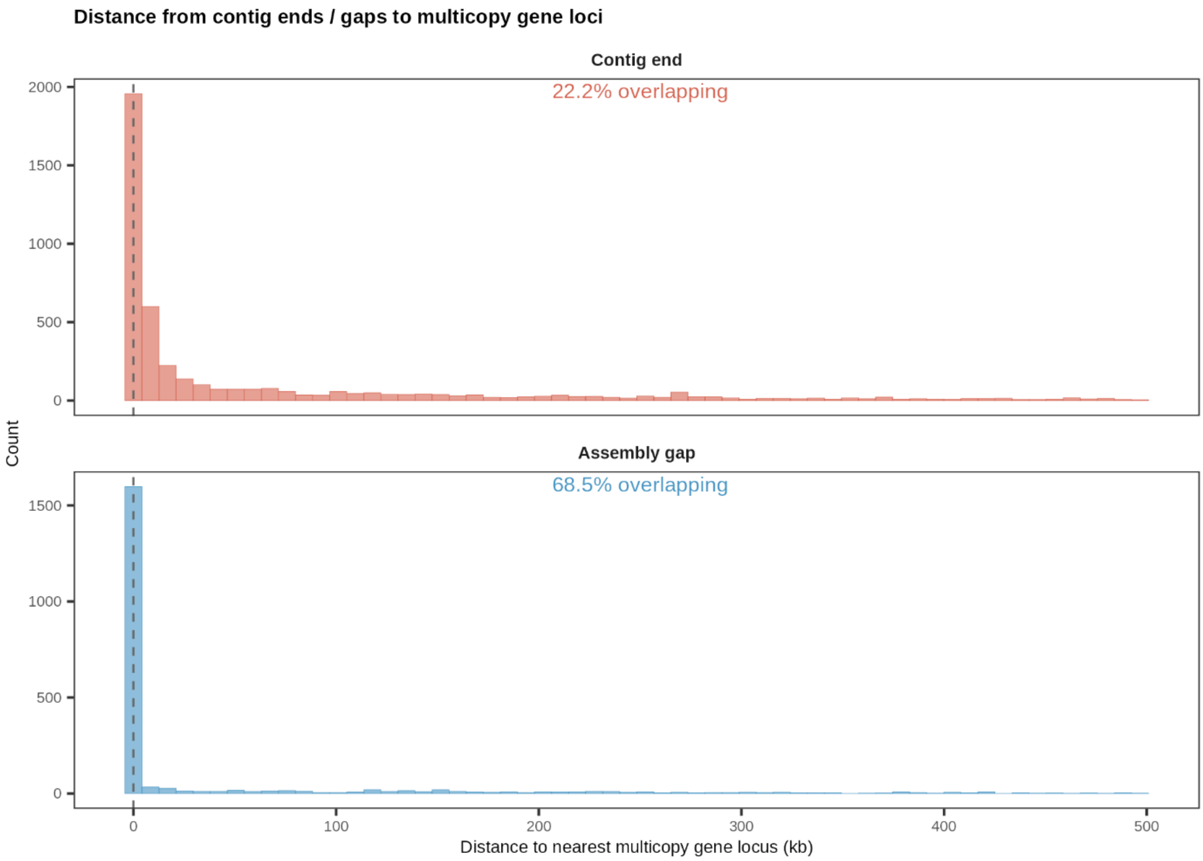
Supplementary Figure 7. Extended reference-free assembly quality metrics. *Comprehensive comparison of Nanopore-only assemblies (red, left) and randomly selected HPRC assemblies (gray, right). The HG002 assemblies (T2T-ONT and HPRC) are included with a triangle shape. Metrics include T2T contig and scaffold counts, gap statistics, inter- and intra-chromosomal misjoins (paftools misjoin), genome completeness, missing multi-copy genes (MMC), missing single-copy genes (MSC) (paftools asmgene), and reciprocal genome coverage statistics (paftools asmstat) relative to T2T-CHM13 v2.*



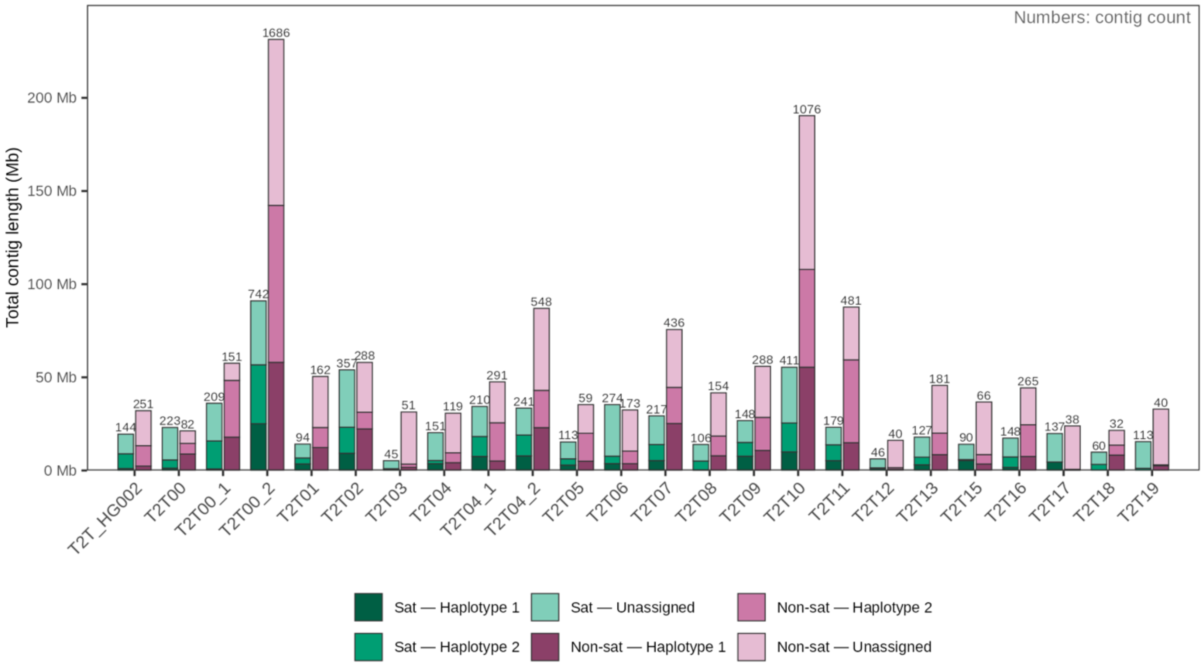
Supplementary Figure 8. Flagger-based detection of potential assembly issues. *Total ~~number~~ length of potential assembly issues identified in both haplotypes using Flagger based on read realignment coverage patterns.*



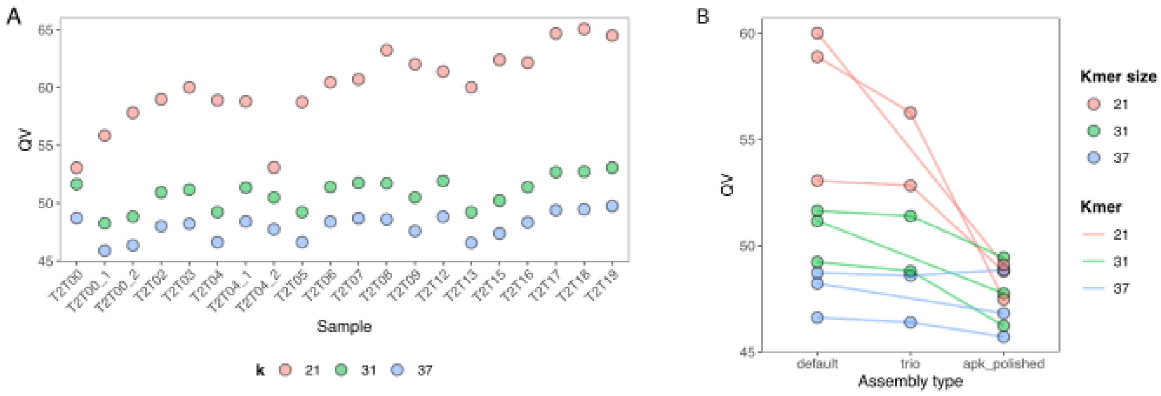
Supplementary Figure 9. Genomic distribution of potential assembly issues. *Combined localization of candidate assembly issues detected by Flagger and NucFlag. Coordinates were lifted to T2T-CHM13 v2 reference positions. To avoid overrepresentation by low-quality samples, counts were normalized across assemblies.*



Supplementary Figure 10. Distance from contig ends and assembly gaps to multi-copy gene loci. *Distance from each contig end (top) and gap (bottom) to the nearest multi-copy gene locus on CHM13v2, up to 500 kb. 22.2% of contig ends and 68.5% of gaps overlap a multicopy gene locus directly. 41% of contig ends are found within 10kb of multi-copy genes. Multi-copy gene loci were annotated using CenSat annotation.*



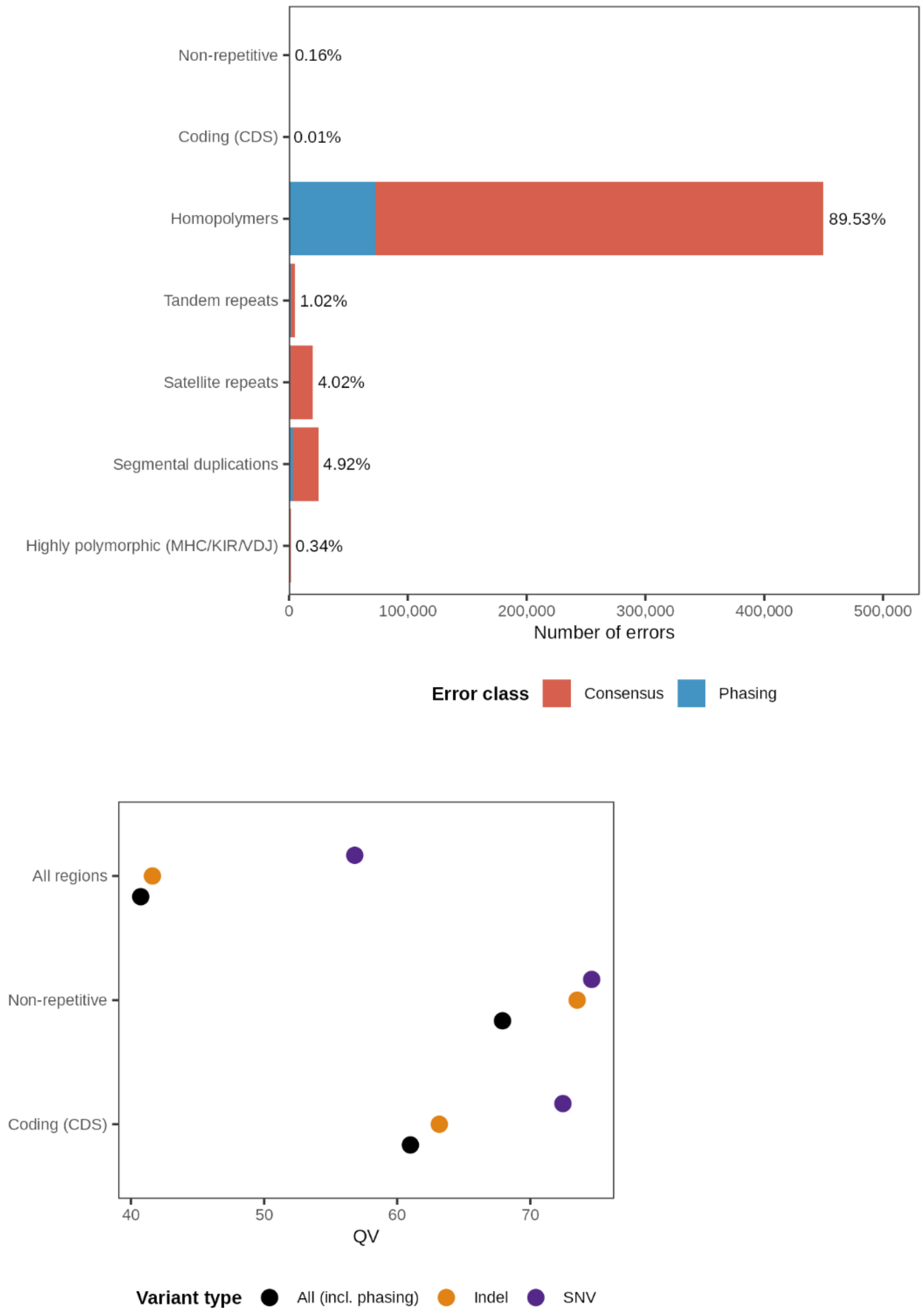
Supplementary Figure 11. Satellite and non-satellite composition of small contigs. *Total length of small contigs (10 kb to 10 Mb) per assembly, split by satellite content (at least 50% aligned span overlapping CenSat satellite counts as satellite) and haplotype. Numbers above the bars give contig counts. Non-satellite contigs are 58.2% of resolved small contigs.*



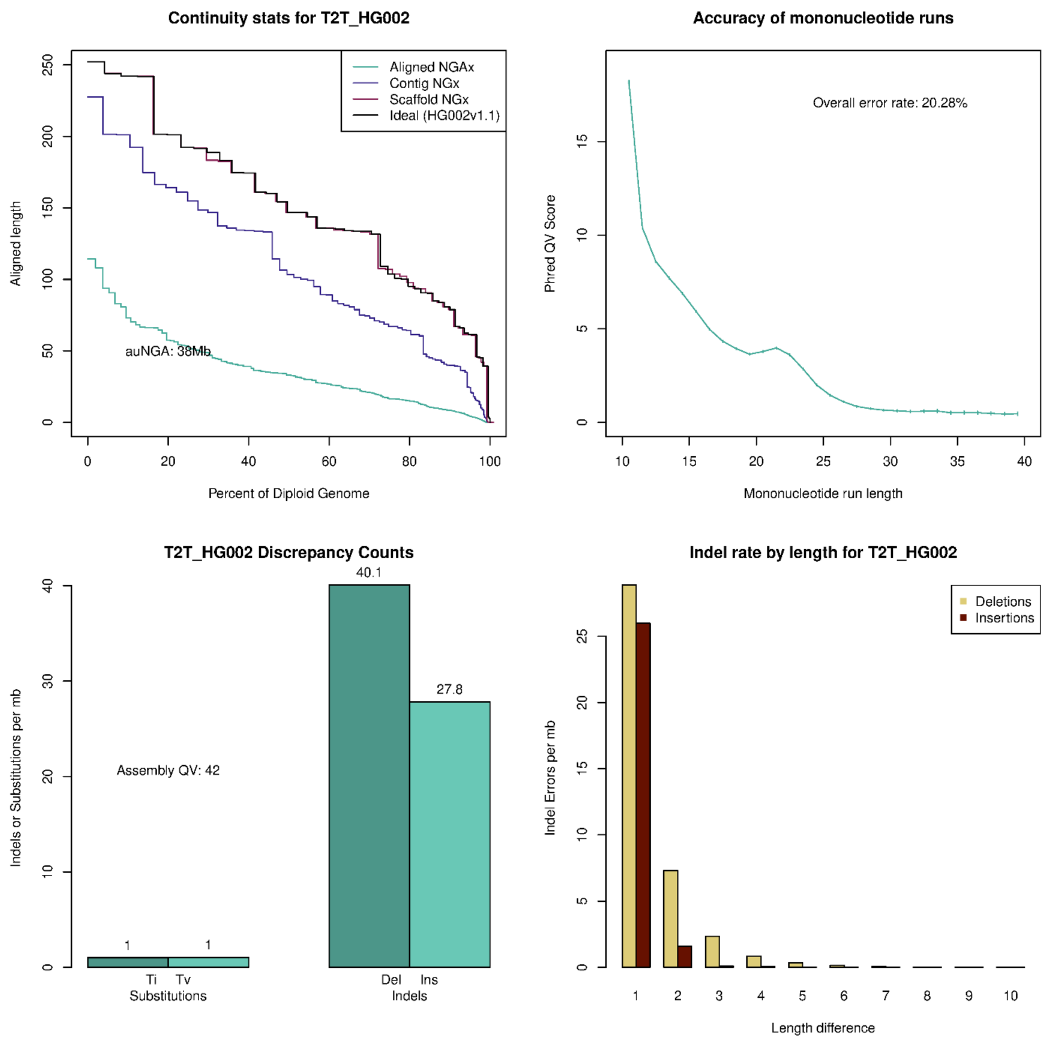
Supplementary Figure 12. Consensus quality values (QV) across assembly strategies and samples. (A) *Merqury-derived QV values calculated using Illumina NovaSeq X Plus reads for k-mer sizes 21, 31, and 37 across all assemblies.* ***(B)*** *Comparison of QV values for different assembly strategies: Verkko with Pore-C phasing (default), Verkko with trio phasing, and Verkko with Pore-C followed by APK polishing.*

A

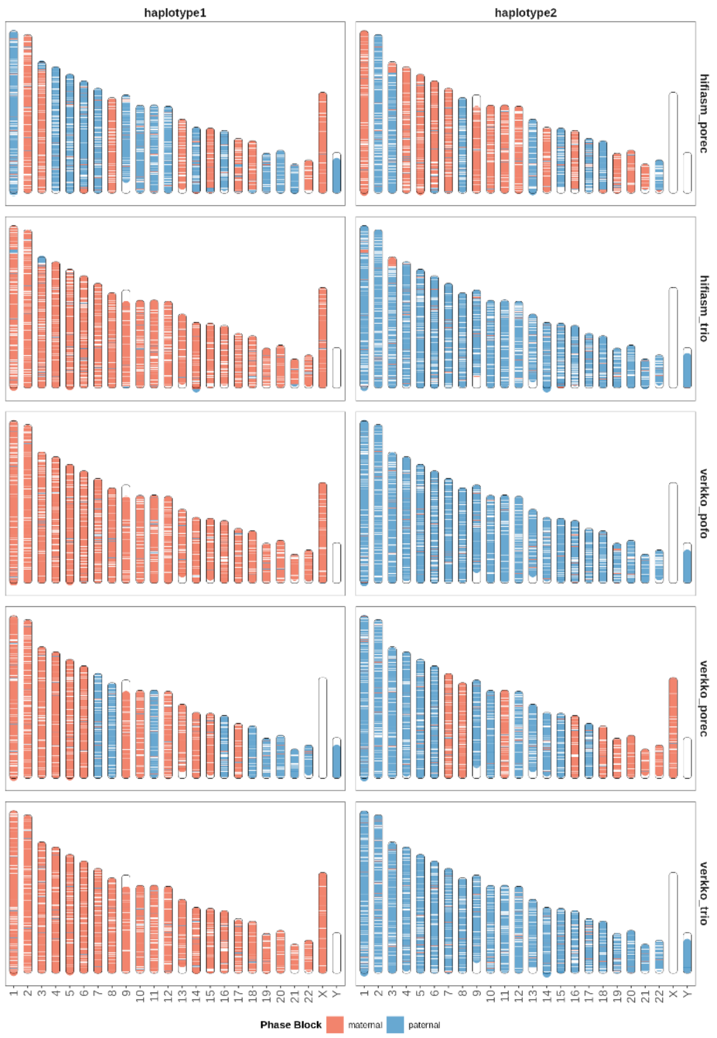
B



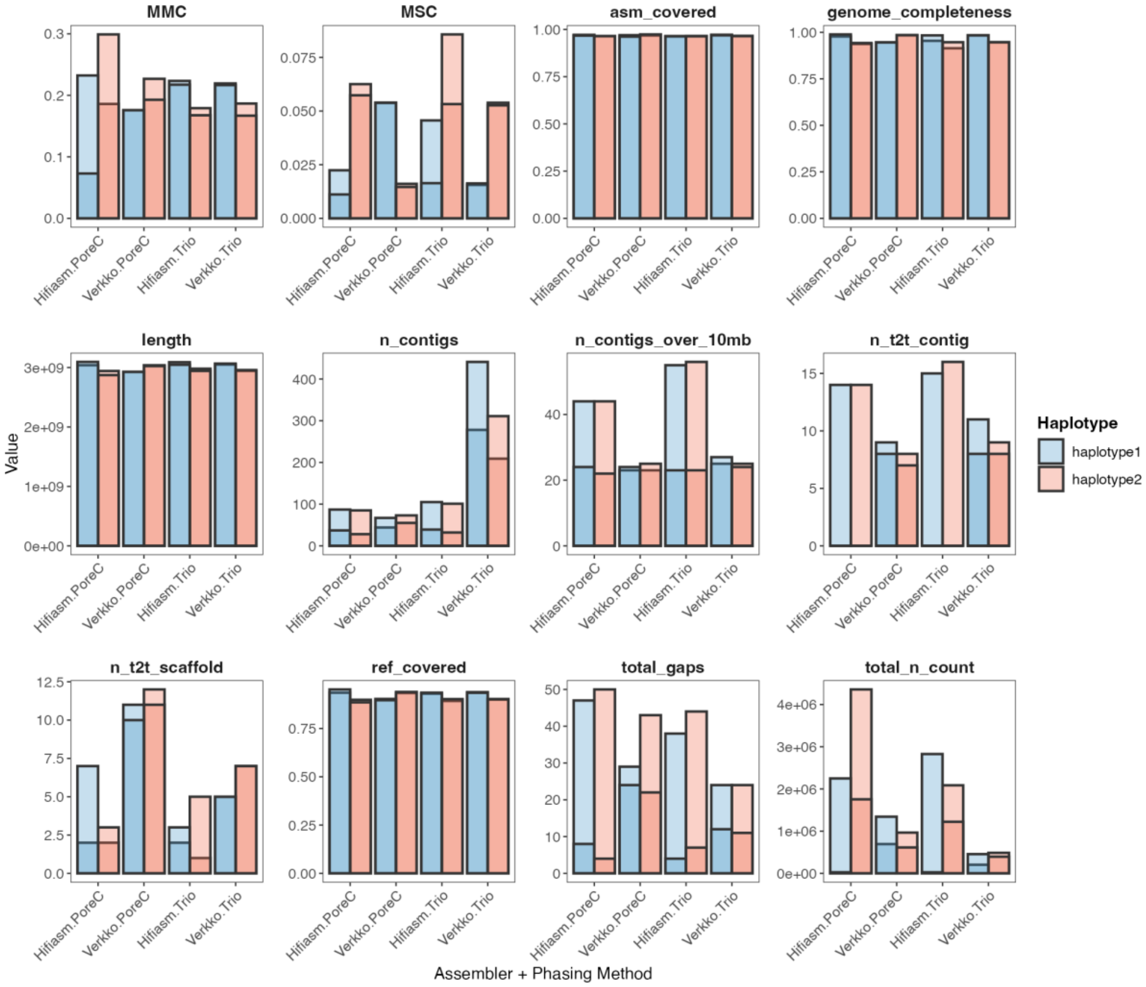
Supplementary Figure 13. Context classification of assembly errors in an independent HG002 assembly. *An HG002 assembly generated with the T2T-ONT workflow (three UL and one Pore-C flow cell) was compared to the GIAB HG002v1.1 benchmark using GQC. Errors were assigned to mutually exclusive categories using the GIAB v4.2 genome stratifications (priority order: coding > highly polymorphic > satellite > segmental duplication > tandem repeat > homopolymer > non-repetitive; see Methods).* ***(A)*** *Number of errors per category, separated into consensus errors (base-level differences from the benchmark) and phasing errors (bases correct for the individual* *but assigned to the incorrect haplotype). Percentages give each category's share of all errors. Homopolymer tracts account for 89.5% of errors, whereas coding and non-repetitive sequence together contribute 0.17%.* ***(B)*** *Assembly quality values by variant type, shown for the three categories in which the number of covered benchmark bases can be determined reliably. QV is not reported for repeat categories because the covered-base denominator is unreliable in those contexts (see Methods). Substitutions (SNV) account for 2.6% of all errors.*



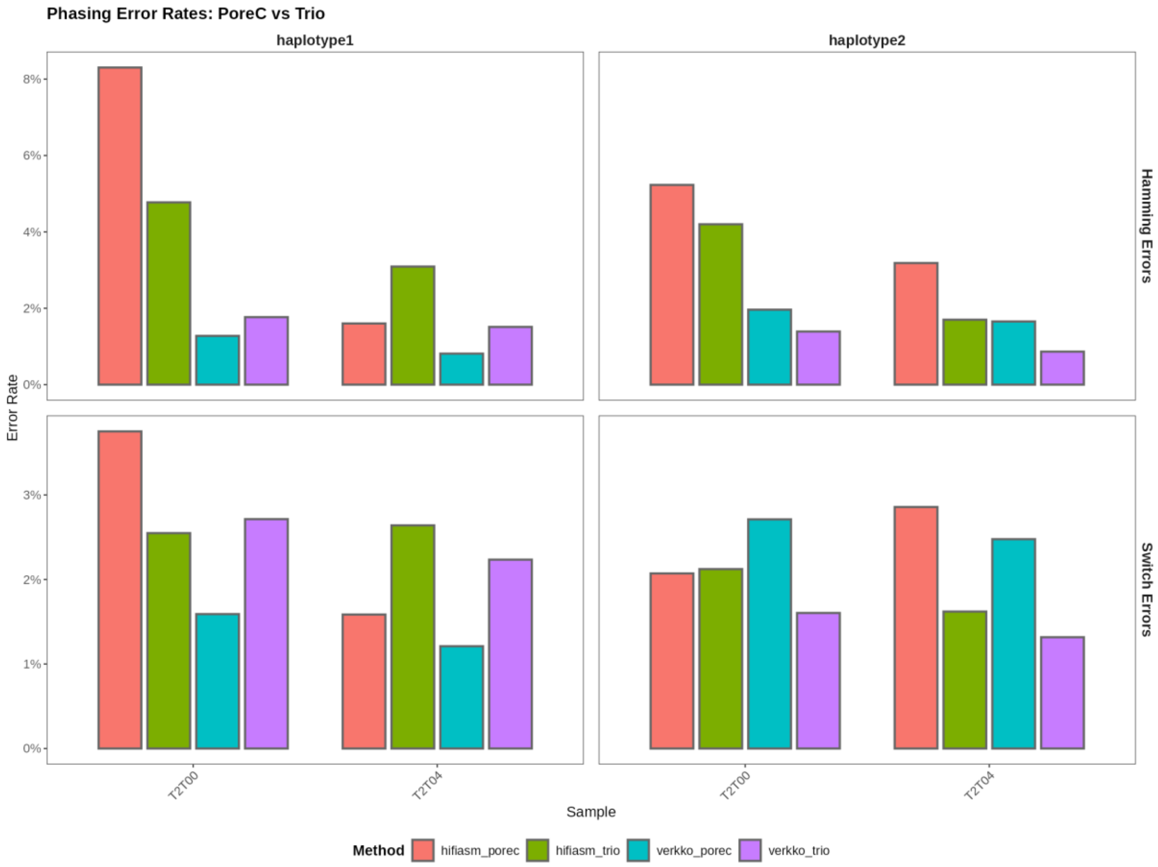
Supplementary Figure 14. Classification of assembly errors in an independent HG002 assembly. *GiaB HG002 reference-based assembly evaluation of our T2T-ONT HG002 assembly was performed using GQC.* ***(A)*** *Continuity statistics: contig NGx (blue), scaffold NGx (red) and aligned NGAx (green) relative to the benchmark assembly (black); auNGA = 38 Mb.* ***(B)*** *Accuracy of mononucleotide runs as a function of run length, expressed as Phred QV. Accuracy declines steeply beyond run lengths of 10 bp; the overall mononucleotide run error rate is 20.3%.* ***(C)*** *Discrepancy counts per megabase, separated into substitutions (transitions, transversions) and indels (deletions, insertions), with the overall assembly QV reported by GQC.* ***(D)*** *Indel error rate per megabase by length difference, separated into deletions and insertions. Single-base length errors dominate, while larger length discrepancies are rare.*



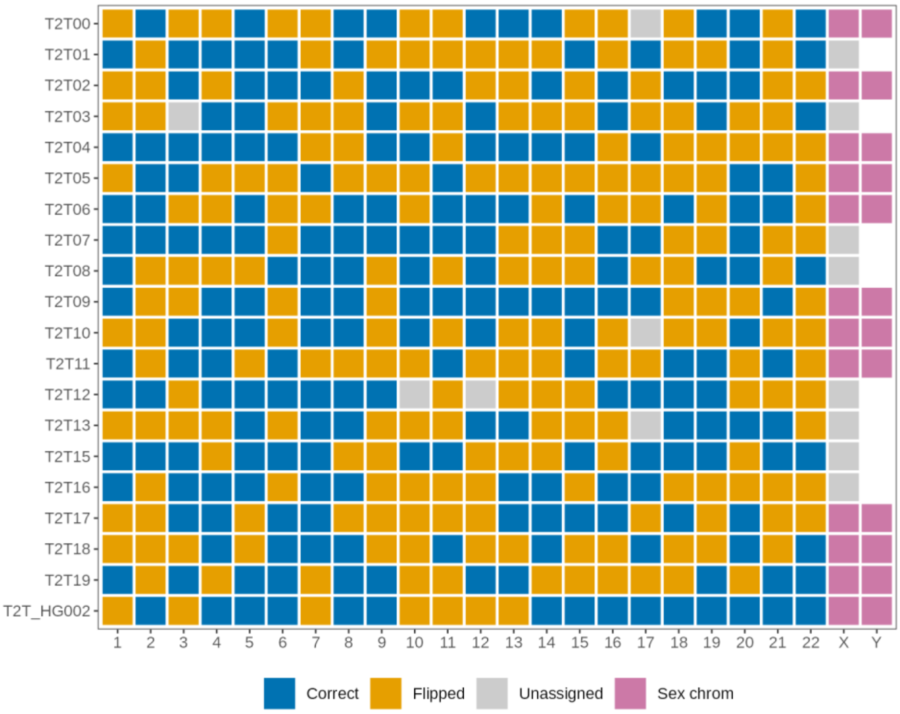
Supplementary Figure 15. Phase block structure and parent-of-origin assessment across assembly methods. *Phase block lengths for sample T2T04 assembly generated using different phasing strategies validated with yak trioeval using identical parental k-mer sets. Blocks are colored by parental origin (maternal, red; paternal, blue). verkko_trio and hifiasm_trio use parental k-mers for phasing. verkko_porec and hifiasm_porec use Pore-C only, for which the assignment of haplotype 1 and 2 to parental origin is arbitrary. verkko_pofo shows the same Pore-C–phased Verkko assembly after methylation-based parent-of-origin assignment with PatMat, which relabels the haplotypes without altering the underlying phasing. Methylation-based assignment recovers the same parental origin as trio-based phasing. hifiasm assemblies show increased large-scale phase switches relative to Verkko.*



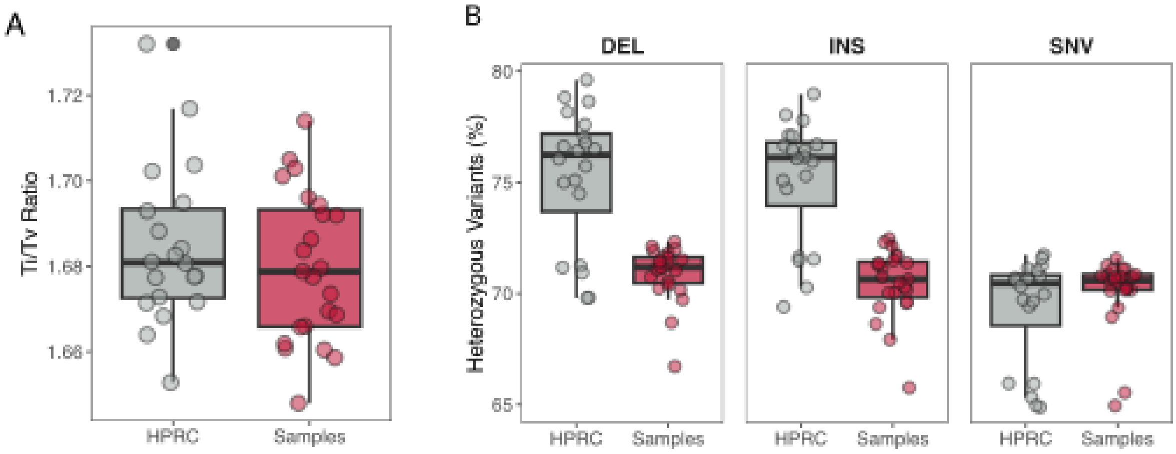
Supplementary Figure 16. QV comparison across assembly and phasing strategies. *Consensus QV values for samples T2T00 and T2T04 generated using four assembly configurations: Verkko with Pore-C phasing, Verkko with trio phasing, hifiasm with Pore-C phasing, and hifiasm with trio phasing.*



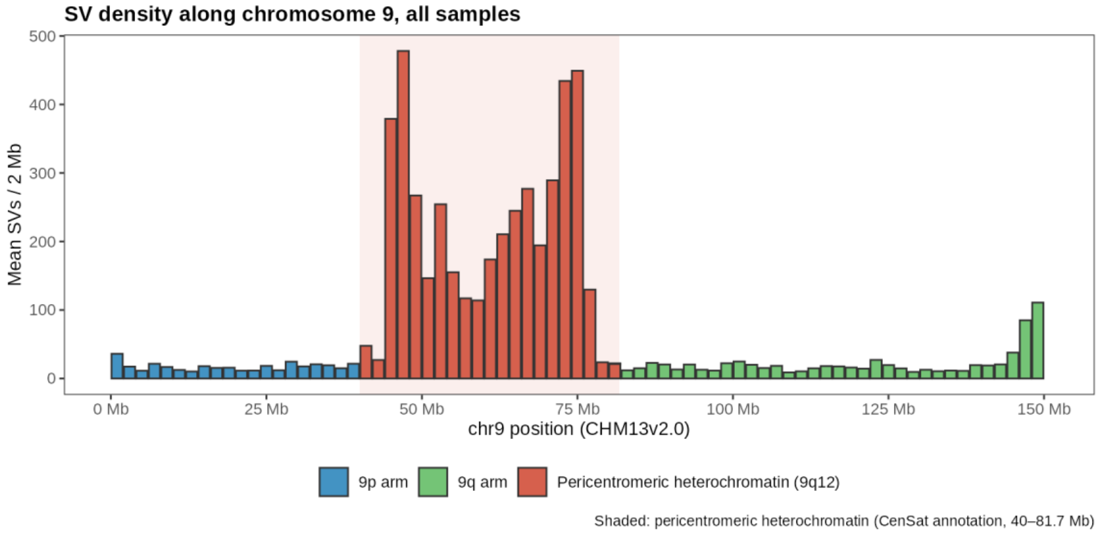
Supplementary Figure 17. Switch and Hamming error rates across phasing methods. *Bar plots of Hamming error rates and switch error rates calculated using whatshap compare for Verkko (trio and Pore-C phasing) and hifiasm (trio and Pore-C phasing). Error rates are shown for both haplotypes in samples T2T00 and T2T04. Verkko demonstrates consistently lower error rates, with comparable performance between Pore-C and trio phasing.*



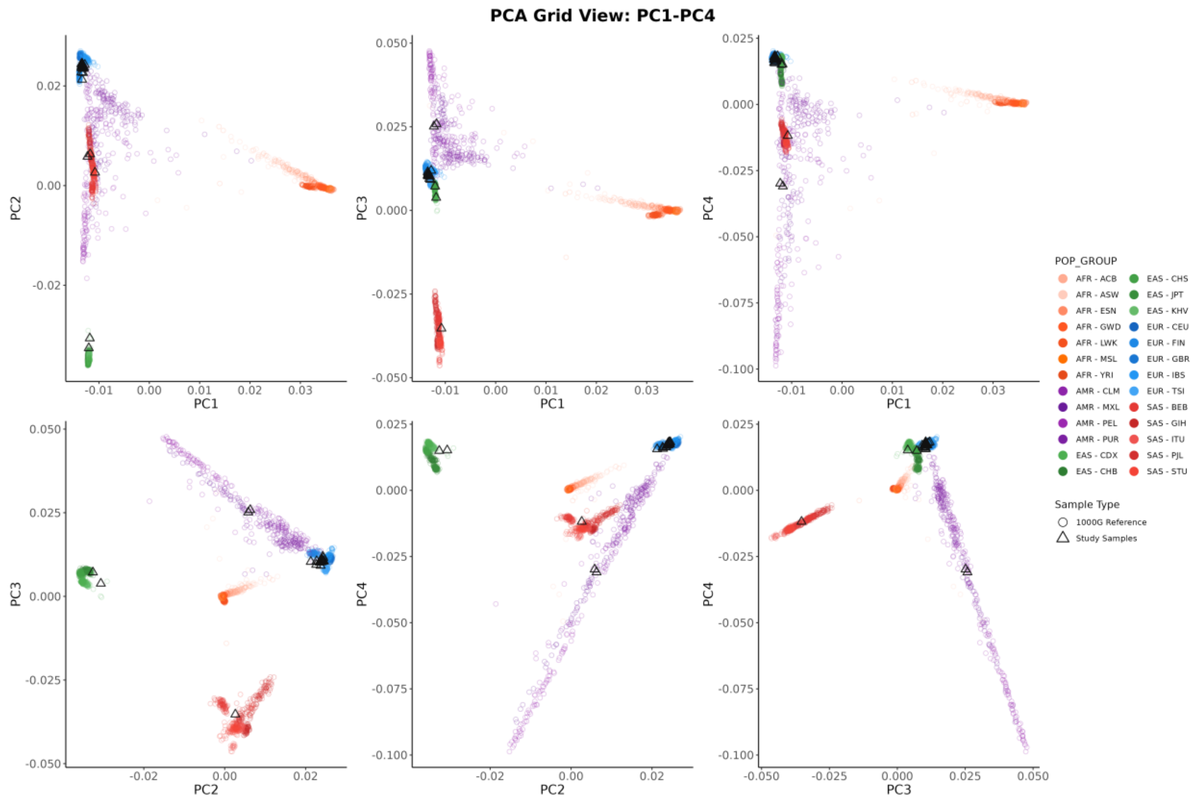
Supplementary Figure 18. Parent-of-origin assignment across the cohort. *Chromosome-level parent-of-origin assignment obtained with PatMat from methylation at imprinted DMRs. For each chromosome, color indicates whether the maternal haplotype corresponds to Verkko haplotype 1 (blue) or haplotype 2 (orange). Since Verkko assigns haplotype labels independently for each chromosome, the distribution of the two categories is arbitrary and does not reflect assignment accuracy. Grey indicates chromosomes for which no confident assignment could be made. Hemizigous sex chromosomes (pink) were excluded from the dataset, also there are no imprinted DMRs annotated on chromosome X as X-chromosome inactivation would in any case dominate the methylation signal. Assignment was obtained for all but six autosomes across the cohort.*



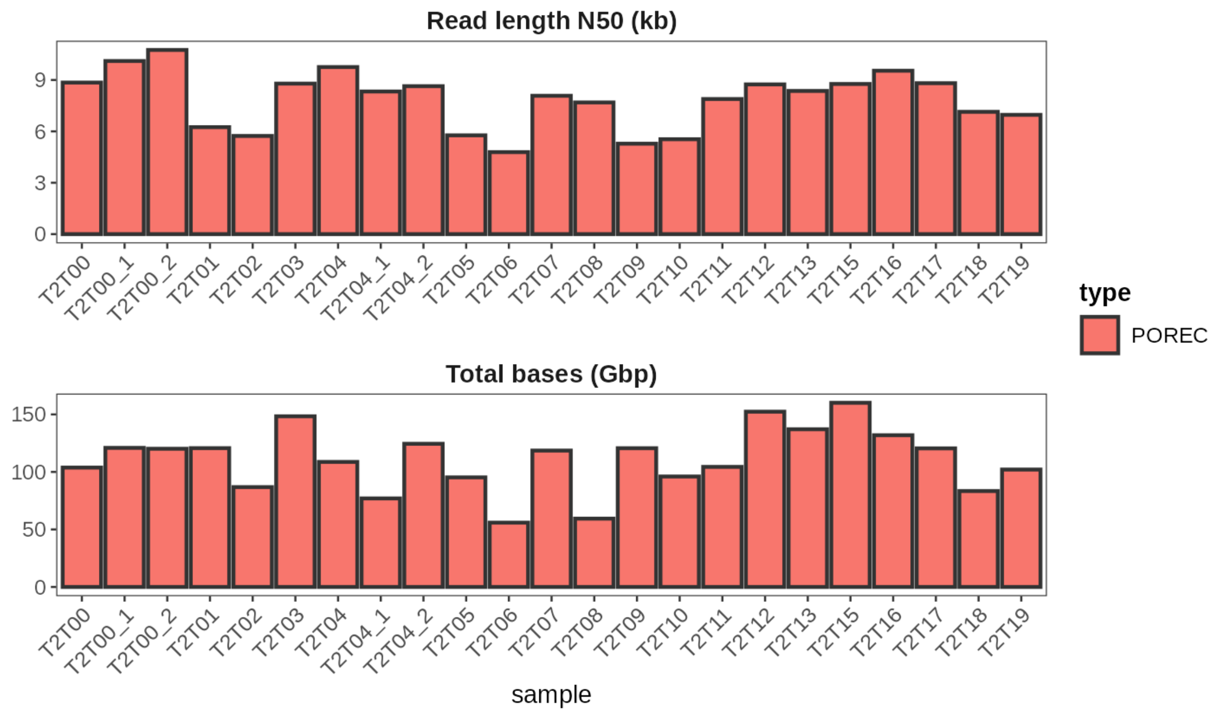
Supplementary Figure 19. Additional small variant characteristics. *Comparison of variant metrics between Nanopore-only assemblies and HPRC assemblies.* ***(A)*** *Transition-to-transversion (Ti/Tv) ratios.* ***(B)*** *Proportion of heterozygous variants per genome.*



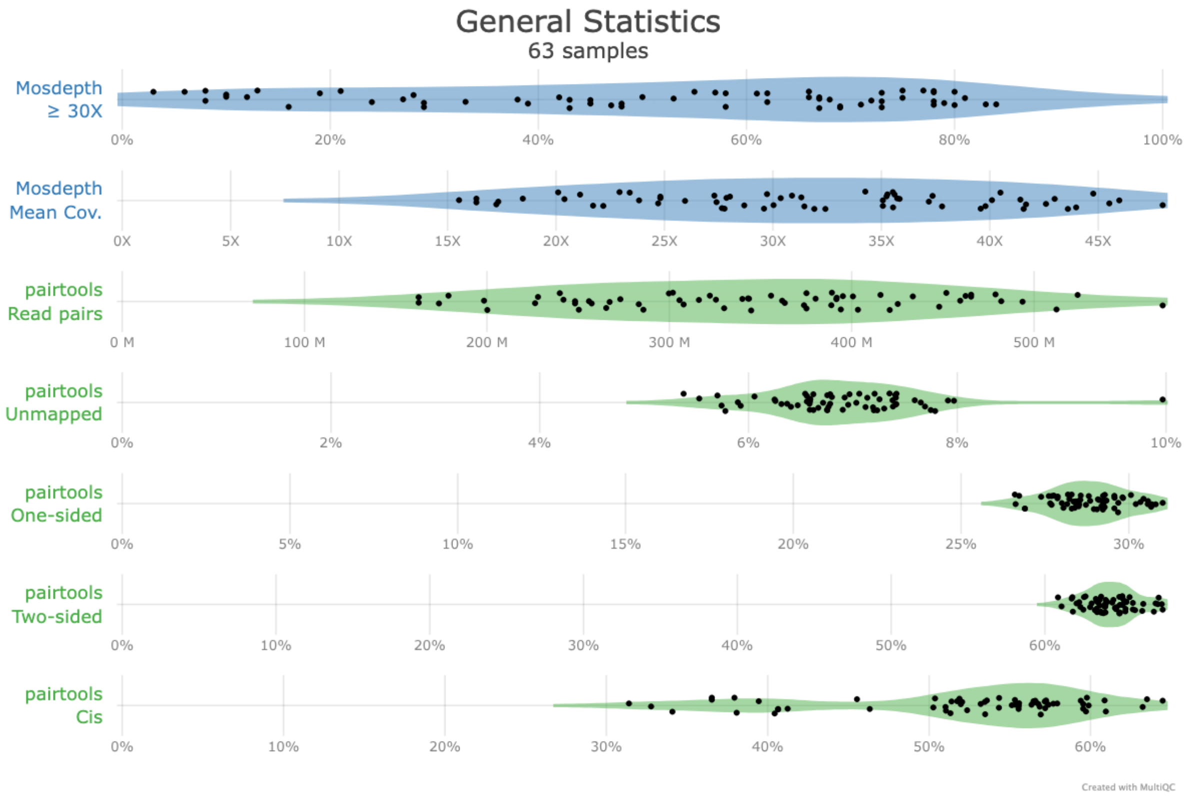
Supplementary Figure 20. High SV density along chromosome 9 is concentrated in the pericentromeric heterochromatic block. *Mean SV calls per 2 Mb bin along chr9 (CHM13v2). The shaded region marks the CenSat centromere and satellite interval (chr9:40.0 to 81.7 Mb). SV calls in this region mostly reflect length divergence between individual heterochromatic arrays and CHM13. SV calling performed using SVIM-ASM.*



Supplementary Figure 21. Principal component analysis of study cohort. *Principal component analysis of phased small variants merged with the T2T-lifted 1000 Genomes reference panel. Plots show pairwise combinations of the first four principal components. Colors represent continental groups.*



Supplementary Figure 22. Pore-C sequencing statistics. *Read statistics for Pore-C datasets used for phasing and scaffolding. For assembly, a single Pore-C flow cell per sample was used. Metrics include total bases, read length distribution, and contact yield.*



Supplementary Figure 23. Quality metrics for individual Pore-C flow cells. *Quality control metrics generated by the wf-pore-c pipeline for each Pore-C flow cell. Only reads containing cis contacts were retained for haplotype phasing and downstream 3D genome analysis.*

Supplementary Table 1: Participant metadata. *Sex, age range at sampling date, and country of birth for each participant. Continental group assignments were derived by the authors from the reported country of birth. T2T00_1/T2T00_2 and T2T04_1/T2T04_2 denote the parents of the two trios.*

Sample	Sex	Age group	Birth country	Continental group
T2T00	Male	65+	Germany	Europe
T2T00_1	Male	65+	Germany	Europe
T2T00_2	Female	65+	Germany	Europe
T2T01	Female	31-64	Russia	Europe
T2T02	Male	31-64	Austria	Europe
T2T03	Female	31-64	Spain	Europe
T2T04	Male	31-64	Germany	Europe
T2T04_1	Male	65+	Germany	Europe
T2T04_2	Female	65+	Germany	Europe
T2T05	Male	31-64	France	Europe
T2T06	Male	31-64	Russia	Europe
T2T07	Female	31-64	Chile	Latin America
T2T08	Female	31-64	India	South Asia
T2T09	Male	31-64	Mexico	Latin America
T2T10	Female	31-64	Poland	Europe
T2T11	Male	31-64	Germany	Europe
T2T12	Female	31-64	Germany	Europe
T2T13	Female	18-30	Taiwan	East Asia
T2T15	Female	18-30	Croatia	Europe
T2T16	Female	31-64	Russia	Europe
T2T17	Male	18-30	Germany	Europe
T2T18	Male	18-30	Germany	Europe
T2T19	Male	31-64	Vietnam	East Asia

Supplementary Table 2. *Number of ONT flow cells generated and used for each genome assembly. Three UL and a single Pore-C flow cell were used for assembly and phasing; additional Pore-C flow cells were sequenced for deep chromatin contact analyses only. In addition, sequencing yield in gigabases, read length N50, and the fraction of reads longer than 100 kb are shown for the uncorrected and unfiltered UL reads, combined per sample. Parental samples of the trios include an additional non-UL (LSK) flow cell in the high-quality dataset. Assembly quality is summarized by the number of assembled T2T contigs and scaffolds, and by the number of complete and correctly assembled centromeres detected by CenMap.*

Sample	Flowcells			Raw UL reads			Assembly QC			
	UL	PoreC	LSK	Yield	N50	Frac	T2T contigs	T2T scaff.	Full cen.	QV
	used/gen	used/gen	n	Gb	kb	>100 kb				
T2T00	3/3	1/4	0	325	48.0	20%	17	23	31	53.1
T2T00_1	3/3	1/2	1	237	67.7	28%	14	22	30	55.8
T2T00_2	3/3	1/2	1	139	77.2	35%	1	18	8	57.8
T2T01	4/4	1/2	0	173	84.8	41%	18	18	14	-
T2T02	4/4	1/2	0	192	100.1	48%	14	21	6	59.0
T2T03	3/3	1/2	0	180	129.3	63%	28	11	20	60.0
T2T04	6/6	1/5	0	292	83.1	37%	15	21	5	58.9
T2T04_1	4/4	1/2	1	169	95.4	45%	5	31	5	58.8
T2T04_2	5/5	1/2	1	159	94.7	45%	3	25	2	53.1
T2T05	3/3	1/2	0	238	74.0	33%	29	11	29	58.7
T2T06	3/3	1/2	0	205	66.9	29%	18	20	20	60.4
T2T07	3/3	1/2	0	201	60.3	23%	3	27	7	60.7
T2T08	3/3	1/2	0	195	92.9	46%	20	18	15	63.2
T2T09	3/3	1/2	0	159	96.0	48%	8	21	10	62.0
T2T10	4/4	1/2	0	158	95.0	47%	0	27	2	-
T2T11	3/3	1/2	0	171	74.9	33%	5	25	17	-
T2T12	3/3	1/5	0	250	80.2	38%	27	14	35	61.4
T2T13	3/3	1/5	0	220	81.1	39%	17	20	20	60.0
T2T15	3/3	1/2	0	247	81.5	38%	22	14	34	62.4
T2T16	3/3	1/2	0	183	84.0	40%	8	30	18	62.1

Sample	Flowcells			Raw UL reads			Assembly QC			
	UL	PoreC	LSK	Yield	N50	Frac	T2T contigs	T2T scaff.	Full cen.	QV
	used/gen	used/gen	n	Gb	kb	>100 kb				
T2T17	3/3	1/2	0	261	79.8	37%	31	8	42	64.7
T2T18	3/3	1/2	0	285	90.5	44%	28	14	38	65.1
T2T19	3/3	1/2	0	299	81.9	39%	29	7	35	64.5

Supplementary Table 3: Mendelian inheritance evaluation for the two trios. *De novo and switch or flip Mendelian error rates and the derived Mendelian QV, restricted to biallelic heterozygous child sites spanned by exactly two contigs (AD 1,1).*

Sample	**Merqury QV**	**Raw MERR (%)**	**Switch/flip (%)**	**De novo MERR (%)**	**Mendelian QV**
T2T00	45.0	4.01	1.37	2.65	48.8
T2T04	44.3	3.57	1.33	2.24	49.5

Supplementary Table 4: Dip3D haplotype assignment rate per sample. *SNV phased fragments: percentage of fragments directly phased by covering a haplotagged SNV. Imputation phased fragments: percentage of fragments phased after phase imputation with neighbouring tagged fragments. Phased reads per chrom: percentage of reads phased per chromosome, averaged across all autosomes by length weighted mean. Phased_reads global: reads assigned to a haplotype in the global union across all chromosomes. A read is counted as assigned if at least one of its fragments on any chromosome carries a tag. Phased contacts: percentage of all contact pairs in which both fragments are haplotype-assigned (imputed tagging).*

Sample	SNV phased fragments (%)	Imputation phased fragments (%)	Phased reads per chrom (%)	Phased_reads global (%)	Phased_contacts (%)
T2T00	12.9	47.3	21.1	46.4	81.8
T2T01	13.4	44.4	23.5	46.7	72.8
T2T02	12.5	30.6	17	44	69.3
T2T03	13.9	49.1	24.1	51.5	79.6
T2T04	12.7	45.5	19.9	44.7	81.6
T2T05	13.3	44.2	23.4	44.3	71.8
T2T06	12.4	27.6	16.2	43.3	67.1
T2T07	13.6	49.6	24.5	52.1	78.5
T2T08	12.4	37	17.4	45.4	78.4
T2T09	12.5	29.7	16.4	44.9	72.5
T2T10	12.7	32.7	17.3	43.3	73
T2T11	13.3	48.1	23.3	51.4	78.9
T2T12	13.6	51	24.6	48.9	80.7
T2T13	12.4	45.3	21.2	43.4	77.3
T2T15	13.4	48.1	22.8	51.1	80.1
T2T16	13.5	55.5	26.5	54.9	82.8
T2T17	13.5	48.7	23.6	50.3	79.4
T2T18	13.2	43.7	21.3	39	77
T2T19	12.1	38.8	18.4	43.5	76.7

Supplementary Table 5. Selection of HPRC release 2 samples used for comparison. *Reference sample HG002 was manually included, the remaining 19 samples were sampled from the prerelease v0.6.1 index (Feb 15 2025, Commit #e0c9453) of the HPRC dataset published at https://github.com/human-pangenomics/hprc_intermediate_assembly using a python script.*

Sample ID	Paternal Haplotype accession ID	Maternal Haplotype accession ID	Assembler	Version	Phasing
HG002	GCA_018852605.3	GCA_018852615.3	Verkko	1.1	trio
HG00272	GCA_044166515.1	GCA_044166665.1	hifiasm	0.19.9	Hi-C
HG005	GCA_018506945.2	GCA_018506965.2	hifiasm	0.19.9	trio
HG00639	GCA_042037765.1	GCA_042037155.1	hifiasm	0.19.9	trio
HG01252	GCA_042038865.1	GCA_042038095.1	hifiasm	0.19.9	trio
HG01361	GCA_018469705.2	GCA_018469685.2	hifiasm	0.19.9	trio
HG02083	GCA_042030135.1	GCA_042030175.1	hifiasm	0.19.7	trio
HG02155	GCA_043304525.1	GCA_043304655.1	hifiasm	0.19.9	Hi-C
HG03834	GCA_041899965.1	GCA_041900115.1	hifiasm	0.19.7	trio
HG04228	GCA_042036925.1	GCA_042036915.1	hifiasm	0.19.9	trio
HG06807	GCA_046332035.1	GCA_046332005.1	Verkko	2	trio
NA18522	GCA_042031325.1	GCA_042030975.1	hifiasm	0.19.9	Hi-C
NA18747	GCA_042031315.1	GCA_042031405.1	hifiasm	0.19.9	Hi-C
NA18948	GCA_046332325.1	GCA_046332305.1	hifiasm	0.19.8	Hi-C
NA19043	GCA_042031205.1	GCA_042030595.1	hifiasm	0.19.9	Hi-C
NA20503	GCA_046629205.1	GCA_046629565.1	Verkko	2.2.1	Hi-C
NA20762	GCA_046629215.1	GCA_046629575.1	Verkko	2.2.1	Hi-C
NA20806	GCA_046629235.1	GCA_046629225.1	Verkko	2.2.1	Hi-C
NA20809	GCA_044166435.1	GCA_044166615.1	hifiasm	0.19.9	Hi-C
NA20827	GCA_046629585.1	GCA_046629265.1	Verkko	2.2.1	Hi-C